Packet Tracer – Configuring VPN Transport Mode

Addressing Table

Device	Private IP Address	Public IP Address	Subnet Mask	Site
Private_FTP server	10.44.2.254	N/A	255.255.255.0	Gotham Healthcare Branch
Public_FTP server	10.44.2.253	209.165.201.20	255.255.255.0	Gotham Healthcare Branch
Branch_Router	N/A	209.165.201.19	255.255.255.248	Gotham Healthcare Branch
Phil's computer	10.44.0.2	N/A	255.255.255.0	Metropolis Bank HQ

Objectives

Part 1: Sending Unencrypted FTP Traffic

Part 2: Configuring the VPN Client within Metropolis

Part 3: Sending Encrypted FTP Traffic

Background

In this activity, you will observe the transfer of unencrypted FTP traffic between a client and a remote site. You will then configure a VPN client to connect to the Gotham Healthcare Branch site and send encrypted FTP traffic. The IP addressing, network configuration, and service configurations are already complete. You will use a client device within Metropolis Bank HQ to transfer unencrypted and encrypted FTP data.

Part 1: Sending Unencrypted FTP Traffic

Step 1: Access the Cyber Criminals Sniffer.

a. Click the **Cyber Criminals Sniffer** and click the **GUI** tab.

b. Click the **Clear** button to remove any possible traffic entries viewed by the sniffer.

c. Minimize the **Cyber Criminals Sniffer**.

Step 2: Connect to the Public_FTP server using an insecure FTP connection.

a. Click the **Metropolis Bank HQ** site and click **Phil's** laptop.

b. Click the **Desktop** tab and click on **Command Prompt**.

c. Use the **ipconfig** command to view the current IP address of **Phil's** computer.

d. Connect to the **Public_FTP** server at **Gotham Healthcare Branch**by entering **ftp 209.165.201.20** in the command prompt.

e. Enter the username of **cisco** and password of **publickey** to login to the **Public_FTP** server.

f. Use the **put** command to upload the file **PublicInfo.txt** file to the **Public_FTP** server.

Step 3: View the traffic on the Cyber Criminals Sniffer.

a. Maximize the **Cyber Criminals Sniffer** that was previously minimized.

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b. Click the **FTP** messages displayed on the sniffer and scroll to the bottom of each one.

What information is displayed in clear text?

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c. Type **quit** to exit **Public_FTP** server.

Part 2: Configuring the VPN Client on Phil's Computer

a. From **Phil's** computer, use the **ping** command and target the IP address of the **Branch_Router**. The first few pings may timeout. Enter the **ping** to get four successful pings.

b. On the **Desktop** tab, click on **VPN**

c. Within the **VPN Configuration** window, enter the following settings:

GroupName: .....**VPNGROUP**

Group Key:.....**123**

Host IP (Server IP): ..**209.165.201.19**

Username: .....**phil**

Password: **cisco123**

d. Click **Connect** and Click **OK** on the next window.

What is the Client IP for the client-to-site VPN connection?

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Part 3: Sending Encrypted FTP Traffic

Step 1: View the current IP addressing on Phil's computer.

a. Within the **Metropolis Bank HQ** site, click **Phil's** computer.

b. Click the **Desktop** tab and click on **Command Prompt**.

c. Use the **ipconfig** command to view the current IP address of **Phil's** PC.

What extra IP address is now shown that was not shown before in Part 1 Step 2c?

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Step 2: Send encrypted FTP traffic from Phil's computer to the Private_FTP server.

a. Connect to the **Private_FTP** server at **Gotham Healthcare Branch**by entering**ftp 10.44.2.254** in the commandprompt.

b. Enter the username of **cisco** and password of **secretkey** to login to the **Private_FTP** server.

c. Upload the file **PrivateInfo.txt** file to the **Private_FTP** server.

Step 3: View the traffic on the Cyber Criminals Sniffer

a. Maximize the **Cyber Criminals Sniffer** that was previously minimized.

b. Click the **FTP** messages displayed on the sniffer.

Are there any FTP messages displaying the password of internal or the file upload of PrivateInfo.txt? Explain.

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Suggested Scoring Rubric

Activity Section	Question Location	Possible Points	Earned Points
Part 1: SendingUnencrypted FTP Traffic	Step 3	20	
Part 2: Configure the VPN Client on Phil's Computer	Step 1	10	
Part 3: Send Encrypted FTP Traffic	Step 1	10	
	Step 3	20	
Questions		**60**	
Packet Tracer Score		**40**	
Total Score		**100**	

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